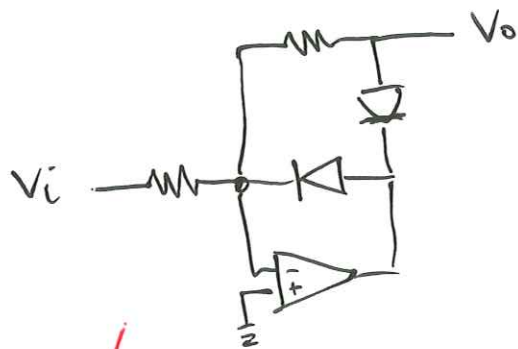
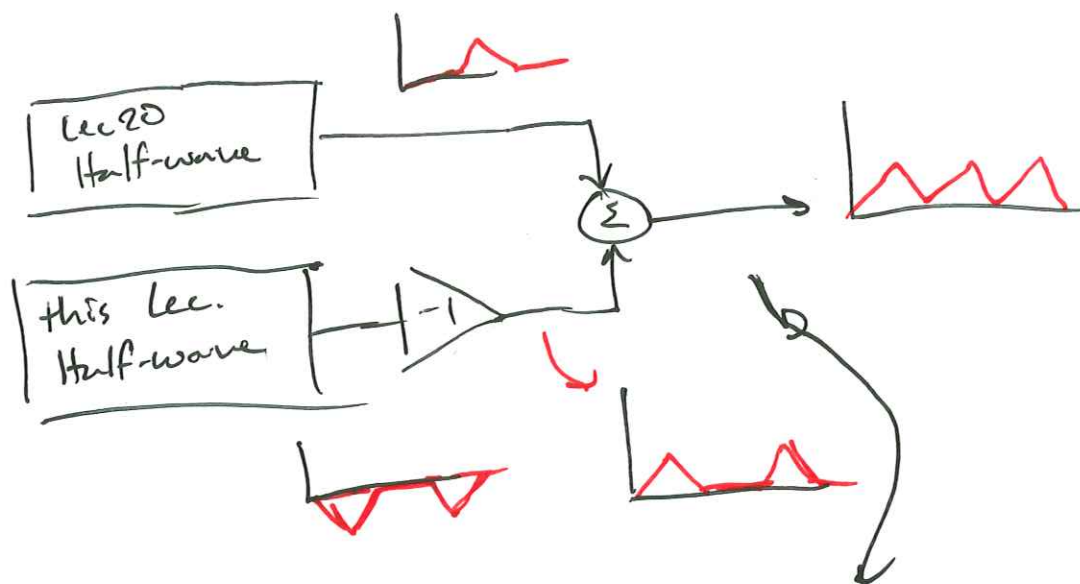
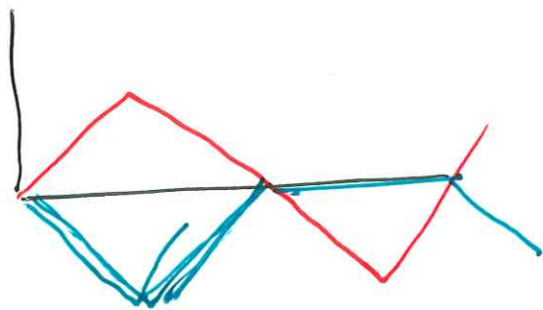




last lecture - ~~the~~ rectifying negative wave
to positive output

→ rectifying positive wave to negative output.

Build a full-wave rectifier:



Better full wave rectifiers --



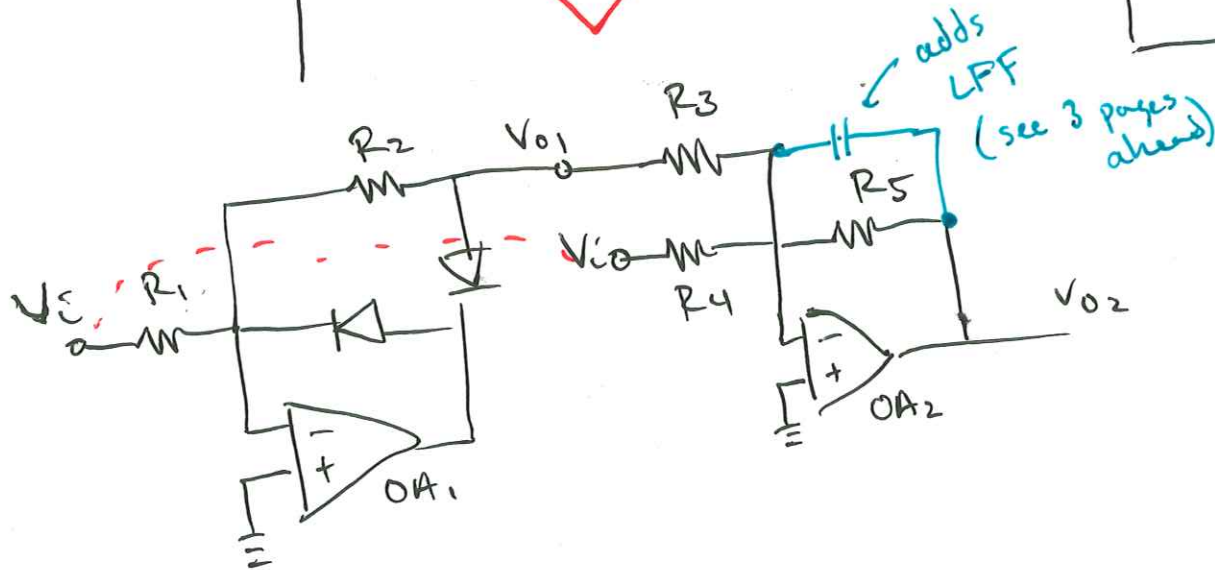
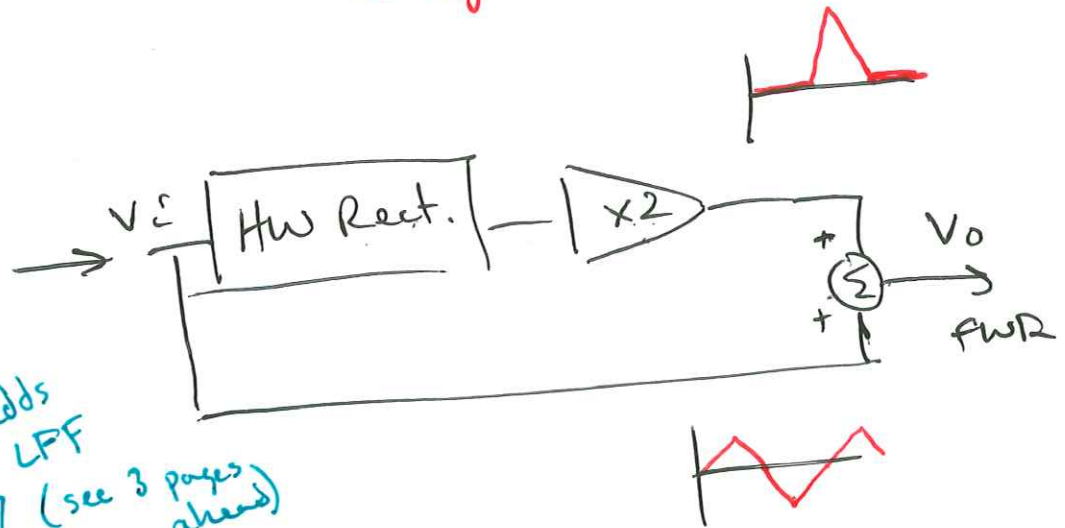
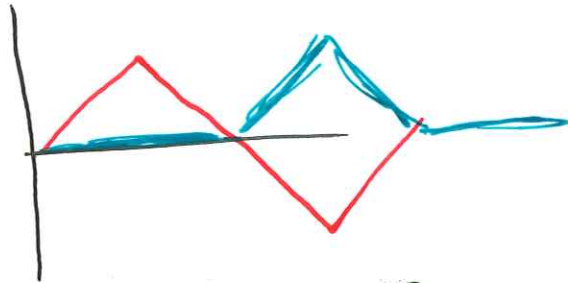
$$V_o = \begin{cases} -V_i & V_i \leq 0 \\ V_i & V_i > 0 \end{cases}$$

No voltage drops!
ideal FWR

Another FWR ...

Assume perfect wave (No DC, negative side == -positive side)
 → symmetric!

Half wave!



$$V_{02} = -R_5 \left(\frac{V_{01}}{R_3} + \frac{V_c}{R_4} \right)$$

$$R_5 = R_4 = 2R_3$$

$$\rightarrow V_{02} = -2V_{01} - V_c$$

$$V_{02} = \begin{cases} -V_c & V_c \leq 0 \\ V_c & V_c > 0 \end{cases}$$

$$V_{02} = |V_c|$$

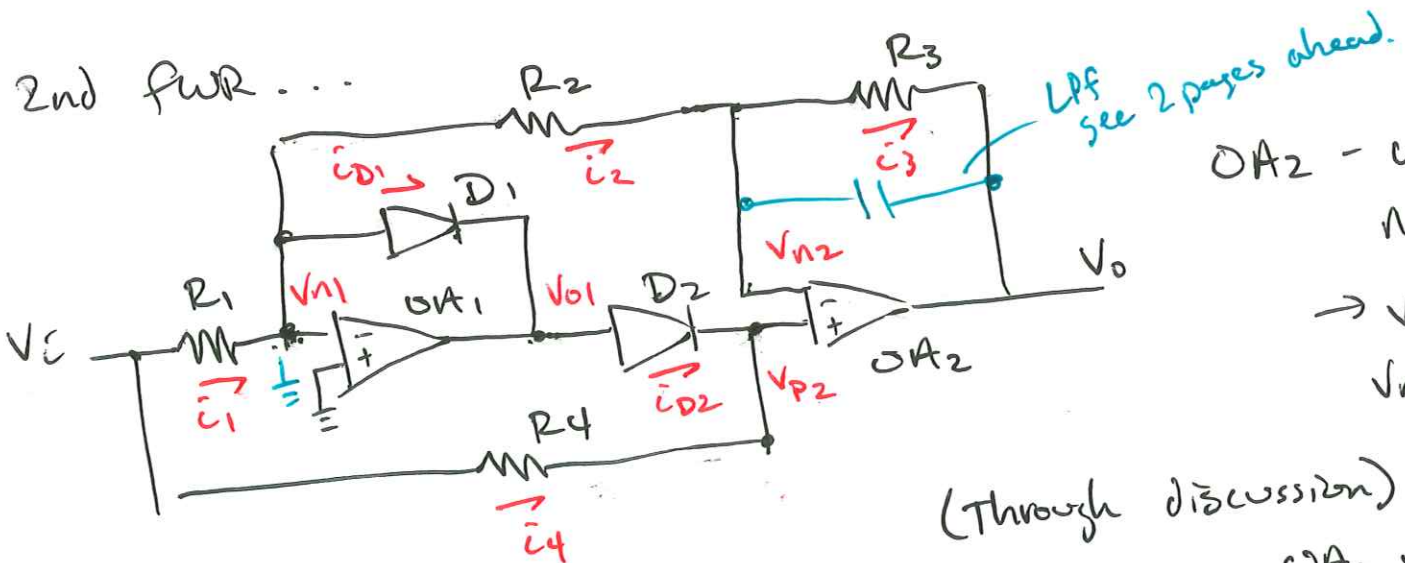
Let's say $R_1 = R_2$

$$V_{01} = \begin{cases} -V_c & V_c \geq 0 \\ 0 & V_c < 0 \end{cases}$$

there is a loading issue!
 OA2 will affect V_{01} !

to avoid! $R_1 = R_2 \ll R_5 = R_4 = 2R_3$

OR add buffer between
 CR... different circuit entirely



$$V_i > 0: V_{n1} = 0,$$

$$\therefore \bar{i}_1 > 0$$

$$D_1 \text{ must be on} \rightarrow V_{o1} = -V_{f1}$$

$$D_2 \text{ must be off} \rightarrow \bar{i}_4 = 0, V_{p2} = V_i$$

$$V_o = \left(1 + \frac{R_3}{R_2}\right) V_i$$

(OA2 operating as non-inv. amp)

$$V_i < 0: V_{n1} = 0$$

$$\therefore \bar{i}_1 < 0, D_1 \text{ must be off}$$

$$\therefore \bar{i}_2 = \bar{i}_1 \rightarrow \frac{V_i - 0}{R_1} = \frac{0 - V_{n2}}{R_2} \rightarrow V_{n2} = -V_i \frac{R_2}{R_1} \Rightarrow V_{n2} > 0$$

$$V_{p2} = V_{n2} = -V_i \frac{R_2}{R_1}$$

$$D_2 \text{ must be on!} \rightarrow V_{o1} = V_{f2} - V_i \frac{R_2}{R_1}, \bar{i}_4 \text{ is } < 0$$

$$V_o = \left(1 + \frac{R_3}{R_2}\right) \left(-V_i \frac{R_2}{R_1}\right)$$

FWR 2 cont.

$$V_o = \left(1 + \frac{R_3}{R_2}\right) \begin{cases} -V_i \frac{R_2}{R_1} & V_i < 0 \\ V_i & V_i > 0 \end{cases}$$

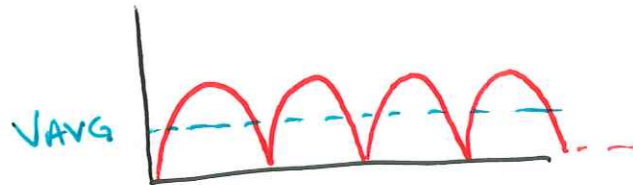
want $R_1 = R_2$ for FWR operation

$$\rightarrow V_o = \left(1 + \frac{R_3}{R_2}\right) |V_i|$$

AC $\rightarrow$ DC conversion

AC $\rightarrow$ V_{RMS} (assume sine wave in...)

FWR:



rectified sine wave now has significant DC component
DC component is average value

for sine waves:

$$V_{AVG} = 0.637 V_{peak}$$

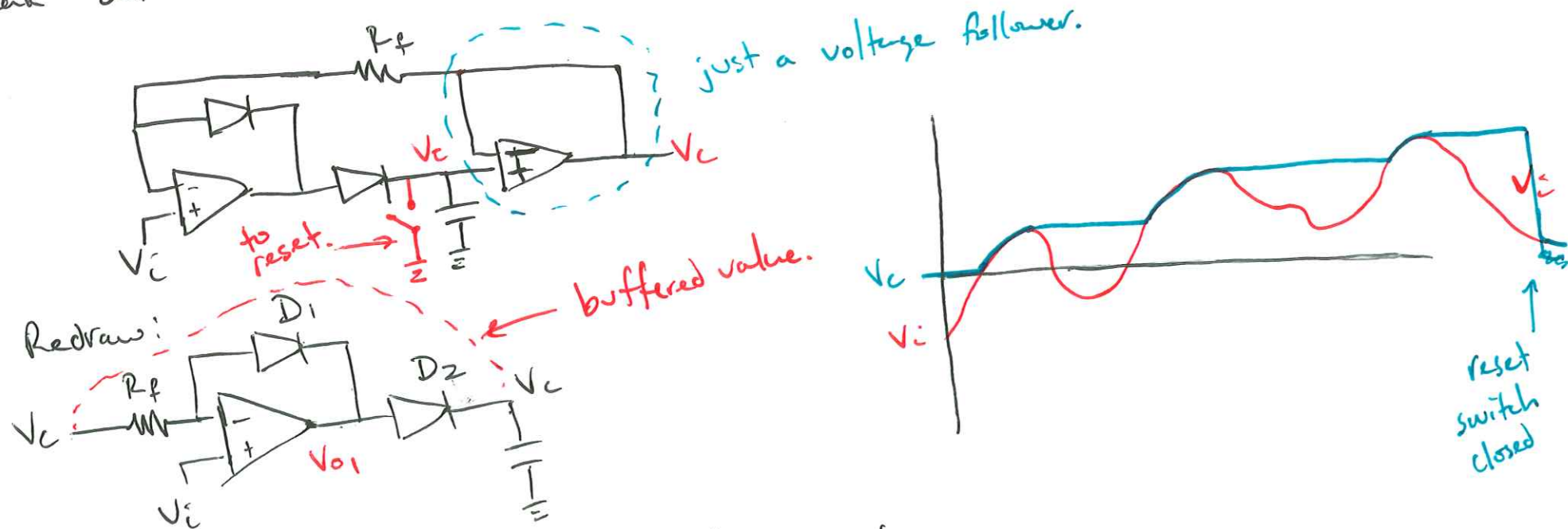
$$V_{RMS} = 0.707 V_{peak} = \frac{0.707}{0.637} V_{AVG} = 1.11 V_{AVG}$$

to get RMS:



$\rightarrow$ Modify previous ^{fwr} circuits to add filtering and gain with no additional opamps.

Peak Detectors $\rightarrow$ measure maximum experienced V_{in}



if $V_i < V_c$: V_{o1} will slew negative $\rightarrow V_{o1}$
 V_{o1} likely not $> V_c$ $\therefore$ D_2 is off. $\rightarrow V_c$ unaffected
 D_1 will turn on $\rightarrow$ Negative feedback active!
 $V_{o1} = V_i$ ~~but V_c~~

$\rightarrow V_i < V_c$, V_c will not change.

if $V_i > V_c$: V_{o1} will slew positive
 D_1 must be off
 D_2 will turn on with $V_{o1} = V_f + V_c$
 $\rightarrow V_c$ will charge to higher voltage.
 Negative feedback through V_c charging!